Bernardo Caldarola

Joint Research Centre – European Commission · Applied Economist
EPSO laureate AD7, EPSO/AD/413/24-9 – Scientific research administrators in social, economic, and political sciences

Edificio Expo, Calle Inca Garcilaso 3, 41092 Seville, Spain | (+34) 608 54 34 22 | bernardo.caldarola@ec.europa.eu
bernacalda.github.io/bernardocaldarola | bernacalda | bernardo-caldarola | B Caldarola

PROFILE

Applied economist with over seven years of quantitative research on innovation, technology and structural change, including two years producing policy analysis for the European Commission. Co-author of DG RTD's SRIP 2026 report (Chapter 10) and of JRC Science for Policy reports on EU capabilities in advanced materials, net-zero technologies and the automotive value chain. Expertise in econometrics and causal inference, economic complexity, and patent, trade, firm and labour-market data; publications in *Research Policy* and other leading journals. Experience with international organisations (UNCTAD, ILO, IFAD). I currently hold academic affiliations with UNU-MERIT (Maastricht University) and Allianz Business School as a fellow of the Productivity Institute (TPI).

PROFESSIONAL EXPERIENCE

Sep 2024 – present — **Economic Complexity Analyst** — Seville, Spain
Joint Research Centre – European Commission

- Quantitative analysis of EU technological capabilities in strategic and emerging technologies, combining patent, trade and labour market data with economic complexity and econometrics
- Co-author of JRC Science for Policy reports and briefs on EU competitiveness and industrial policy, including chapter 10 of DG RTD's *Science, Research and Innovation Performance of the EU (SRIP) 2026* report
- Designed and delivered training in economic complexity for Commission economic analysts at DG RTD (Brussels and Seville, 2025–2026)
- Organising and scientific committees of the CONCORDi conference and the RESKILL workshop

Sep 2023 – Aug 2024 — **Research Fellow** — Maastricht, Netherlands
Maastricht University and UNU-MERIT

- Researcher in the unit of Economics of Innovation and New Technologies and Computational Innovation Lab
- Drafting research projects for national and international funding bodies (EU Horizon, Dutch Research Council, The Productivity Institute)
- Supervision of Master and PhD students in the field of Development Economics and Innovation

May 2022 – Sep 2023 — **Research Fellow** — Rome, Italy
Enrico Fermi Research Centre (CREF)

- Research in Economic Complexity focusing on structural change, employment dynamics, and sustainable transitions
- Drafting research project proposals for national (PRIN) and European (Horizon) funding

Nov 2019 – Feb 2022 — **Research Assistant** — Brighton, UK
University of Sussex – Science Policy Research Unit (SPRU)

- Research Assistant in ICT and Energy Demand, CREDS (Sep 2020 – Feb 2022)
- Research Assistant in Research Infrastructure, RISIS2 (Nov 2019 – Apr 2021)

Nov 2018 – Jul 2019 — **Research Assistant** — Brighton, UK
University of Sussex – Institute of Development Studies (IDS)

- Data cleaning, and econometric analysis of time diary data for the "Growth and Equal Opportunities for Women" (GrOW) programme and for the ActionAid International funded research project "Unpaid care work: Feminist analysis of time diaries"

May 2017 – Sep 2017 — **Evaluation Research Analyst, Intern** — Rome, Italy
International Fund for Agricultural Development (IFAD)

- Research and data collection for the draft of the IFAD's Independent Office of Evaluation Country Strategic Programme Evaluation for Angola
- Research work for the approach paper of a Project Performance Evaluation for an IFAD pastoral water project in rural Chad

EXTERNAL CONSULTANCY

Mar 2024 – Dec 2024 **Consultant** Remote
UN Trade and Development (formerly UNCTAD)
- Empirical analysis of geospatial data to inform Venezuelan industrial policy and diversification strategies

Sep 2023 – Aug 2024 **Consultant** Remote
European Commission's Joint Research Centre – Seville
- Expert support to the SINATRA (Sustainable Innovation and Transition pathways in Rural Areas) exploratory research activity

Mar 2023 – Feb 2024 **Consultant** Remote
Catholic University of the Sacred Heart of Milan & ANSEB
- Analysis of the food voucher industry in Italy

Jul 2021 – Dec 2021 **External Consultant** Remote
International Labour Office (ILO)
- Funded through the "Skills and transitions" research project by the ILO Research Department

Nov 2018 – Mar 2019 **Consultant** Various locations
Palladium International & Institute of Development Studies (IDS)
- Impact evaluation of the DfID funded pilot programme "West African Food Market" to support smallholder farming productivity growth in Ghana, Niger, Nigeria, and Burkina Faso; fieldwork for data collection in Ouagadougou, Burkina Faso

PUBLICATIONS

Reports and Policy Briefs

1. Building the EU technological leadership: Absolute and relative capabilities in advanced materials, with Dario Diodato, Giambattista Albora, Carla Sciarra, Lorenzo Napolitano. *European Commission, Directorate-General for Research and Innovation – Science, Research and Innovation Performance Report (SRIP), 2026*
2. Chinese cars in the EU: Current trends and driving factors, with Biagio Ciuffo, Hugo Confraria, Giuditta De Prato, and others. *European Commission, Joint Research Centre – Science for Policy Brief, 2026*
3. Path to innovation: an Economic Complexity analysis of technological perspectives in the EU, with Giambattista Albora, Florence Benoit, Valentina di Girolamo, Dario Diodato, Lorenzo Napolitano, Carla Sciarra. *European Commission, Joint Research Centre – Science for Policy Report, 2025*
4. Strategic technologies and industrial policies for competitiveness and sustainability, with Carla Sciarra, Clemens Domnick, and Fernando Hervas. *European Commission, Joint Research Centre – Science for Policy Brief, 2025*
5. Global Value Chain analysis of the EU automotive sector under the lens of Economic Complexity, with Lorenzo Cresti, Dario Mazzilli, Lorenzo Napolitano, Aurelio Patelli, Angelica Sbardella, Andrea Tacchella. *European Commission, Joint Research Centre – Science for Policy Brief, 2025*
6. EU competitiveness in net-zero technologies: Insights from patents and economic complexity, with Giambattista Albora and Robert Marschinski. *European Commission, Joint Research Centre – Science for Policy Brief, 2025*

Journal Articles

1. Scaling Up to the Cloud: Cloud Technology Use and Growth Rates in Small and Large Firms, with Luca Fontanelli. **Research Policy**, forthcoming.
2. 3G internet diffusion and secondary education attainment: Evidence of opportunity cost in Viet Nam, with Trang Thi-Pham. **Review of Development Economics**, 2026, 30(2), 941–966
3. Economic complexity and the sustainability transition: A review of data, methods, and literature, with Dario Mazzilli, Lorenzo Napolitano, Aurelio Patelli, and Angelica Sbardella. **Journal of Physics: Complexity**, 2024, 5(2)
4. Does teleworking encourage longer commutes? Evidence from the English National Travel Survey, with Steve Sorrell. **Travel Behaviour and Society**, 2024, 37, 100822
5. Mobile internet, Skills and Structural Transformation in Rwanda, with Marco Grazzi, Martina Occelli, Marco Sanfilippo. **Research Policy**, 2023, 52(10), 104871

Media coverage: Nemo Kennislink 1, Nemo Kennislink 2

6. Do teleworkers travel less? The historical impact of teleworking on English travel, with Steve Sorrell. **Transportation Research Part A: Policy and Practice**, 2022, 159, 282–303
   Media coverage: Forbes, The Telegraph, Daily Mail, and others.

Working Papers

1. Structural Change, Employment, and Inequality in Europe: An Economic Complexity Approach, with Dario Mazzilli, Aurelio Patelli, Angelica Sbardella.
   UNU-MERIT Working Paper Series, arXiv:2410.07906.
2. Mobile Internet Adoption and Inclusive Structural Change: Evidence from Nigerian Non-Farming Enterprises.
   STEG Working Paper.
3. Structural Change(s) in Ghana: A Comparison between the Export, Formal, and Informal Sectors.
   LEM Papers Series, Sant'Anna School of Advanced Studies, Pisa, Italy.
4. Informal and Formal Industrial Co-location and Structural Change in Ghana, with Tommaso Ciarli.
5. Women's depletion risk from time poverty: Evidence from Nepal, India, Rwanda and Tanzania, with Amrita Saha and Deepta Chopra.

Work in Progress (Selected)

1. Causal Inference for Economic Complexity and Relatedness.
   With Giacomo Lo Conte, Dario Diodato, Lorenzo Napolitano, Giambattista Albora, Carla Sciarra.
2. Absolute and relative technological capabilities.
   With Ron Boschma, Dario Diodato, Lorenzo Napolitano, Giambattista Albora, Carla Sciarra.
3. Semantic product space.
   With Nico Sherpa, Dario Diodato, Giambattista Albora, Carla Sciarra.
4. Industrial Policy and New Export Activities: The Role of Existing Capabilities
   With Alessio Bumbea, Emanuele Pugliese, Dario Diodato, Carla Sciarra, Giambattista Albora.
5. AI capabilities and labour market outcomes: Evidence from EU regions.
   With Maddalena Mula, Carla Sciarra, Dario Diodato, Enrique Macias-Fernandez, Pablo Casas, Lorenzo Napolitano, Giambattista Albora.
6. The in-demand skills portfolios of UK and EU regions and their impact on productivity.
   With Tommaso Ciarli, Alexandra Badort, Sidharth Rony.
7. The adoption of 3D printers and its impact on the value chain positioning of French firms.
   With Luca Fontanelli, Kristina McElheran, Elena Verdolini
8. The Hidden Innovators: Uncovering Capabilities and Industrial Dynamics in European Rural Areas.
   With Simone Sasso, Simon Trlifaj, Tommaso Ciarli, Emanuele Pugliese, Alessio Bumbea.

EDITORIAL AND PEER REVIEW ACTIVITIES

Editorial

Coeditor of the Special Issue *Innovation and strategic autonomy for competitiveness*, with Michele Cincera, Helena Lenihan, Hannah Hottenrott, Pietro Moncada-Paternò-Castello, and Francesco Rentocchini. **Economics of Innovation and New Technology**

Peer review

Research Policy (5), Journal of Evolutionary Economics (4), Journal of International Business Economics (4), Structural Change and Economic Dynamics (3), Economics of Innovation and New Technology (2), Journal of Population Economics (1), Economic Geography (1), Regional Studies (1), Spatial Economic Analysis (1), Industry and Innovation (1), Journal of Development Effectiveness (1), African Development Review (1), Asian Development Review (1), B.E. Journal of Economic Analysis and Policy (1), Empirical Economics (1), Transportation Research Part A: Policy and Practice (1), Journal of Transport Geography (1), Journal of Physics: Complexity (1), Manchester School (1), PLOS Climate (1), Scienze Regionali (1).

EDUCATION

Sep 2017 – Jan 2023 **PhD Programme in Science and Technology Policy Studies** Brighton, UK
University of Sussex Business School – Science Policy Research Unit

- Thesis title: "Drivers and Impacts of Structural Change in Sub-Saharan Africa: Evidence from

Households and Firms"
- Evaluation: Pass with minor corrections
- Supervisors: Dr Tommaso Ciarli; Prof Maria Savona
- Examiners: Prof Margaret McMillan (Tufts University – Massachusetts, United States, external); Prof Raphie Kaplinsky (internal)
- Description: Analysis on the drivers and impacts of structural change in Sub-Saharan Africa with a focus on informality, complexity, and digital technologies, using a range of econometric and network techniques.

Sep 2014 – Nov 2016 **Master's Degree in Development and International Cooperation** Lisbon, Portugal
Lisbon School of Economics and Management (ISEG) – ULisboa
- Dissertation title: "Poverty Traps in Southern Rural Mozambique: a Structural Equations Modelling Approach"
- Final grade: 18/20 (1st percentile)
- Erasmus exchange programme at Université Paris X – Nanterre la Défense, Department of Economics (fall term 2015–16)

Sep 2009 – Apr 2013 **BA in Political Science and International Relations** Pavia, Italy
Università degli Studi di Pavia
- Dissertation title: The Role of Microcredit in Economic Growth and Poverty Reduction
- Final grade: 104/110

LANGUAGES AND SKILLS

Languages	Italian (native), English (C2), Spanish (C2), French (B2), Portuguese (B2), Bissau Creole (B1)
Research	Econometrics, causal inference, impact evaluation (DiD, IV, shift-share, synthetic control), geocomputation, network and data science
Software	Coding and data analysis: R, Python, Stata, Gephi, VOSviewer, Cytoscape. Typesetting: LaTeX, RMarkdown, Quarto, Sublime Text
Certificates	2021 — Office for National Statistics (ONS) and UK Data Service (UKDS), Accredited Researcher 2019 — Higher Education Academy (AFHEA), Associate Fellow

TRAINING AND TEACHING ACTIVITIES

Training delivered to the European Commission

2026 **Basic and advanced training in Economic Complexity** – Three-day training delivered to DG RTD · Brussels, BE

2025 **Advanced training in Economic Complexity** – Two-day training delivered to DG RTD · Seville, ES

2025 **Introduction to Economic Complexity** – Two-day training delivered at DG RTD · Brussels, BE

University teaching

2024-2025 **The Economic Rationale of Economic Complexity** – Lecturer, Summer School at UNU-MERIT (2024) and CREF (2025) · Maastricht, NL and Rome, IT

2024 **Evidence-Based Policy Research Methods** – Tutor, UNU-MERIT · Maastricht, NL

2022 **Economics of Innovation** – Postgraduate R workshops, Università LUISS Guido Carli. One term · Rome, IT

2019–21 **Innovation and Economic Development** – Postgraduate seminars, University of Sussex (SPRU). Three terms (one online, one hybrid) · Brighton, UK

2018–19 **Quantitative Methods for Science, Technology and Innovation Studies** – Postgraduate Stata workshops, University of Sussex (SPRU). Two terms (one online) · Brighton, UK

2018–19 **Applied Economic Topics: Economics of Innovation** – Undergraduate seminars, University of Sussex (Department of Economics). Two terms (one online) · Brighton, UK

2018 **Stata Surgery Sessions** – Stata workshops for undergraduate finalists, University of Sussex (Department of Economics). One term · Brighton, UK

2018 **Network Analysis and Infographics** – Postgraduate R Studio workshops, University of Sussex, SPRU. One term · Brighton, UK

2018 **Economic Perspectives on Development** – Postgraduate seminars, Institute of Development Studies. One term · Brighton, UK

2017 **Entrepreneurship Finance and Accounting** – Undergraduate seminars, University of Sussex (Department of Management). One term · Brighton, UK

ORGANISATION OF CONFERENCES AND WORKSHOPS (SINCE 2023)

2027 **11th CONCORDi Conference on Corporate R&D and Innovation** – Academic Conference, European Commission Joint Research Centre · Seville, ES

2026 **3rd REsearch, SKills, Innovation and Labour Lab (RESKILL) Workshop** – Academic Workshop, Joint Research Centre – European Commission · Seville, ES

2025 **2nd REsearch, SKills, Innovation and Labour Lab (RESKILL) Workshop** – Academic Workshop, University of Ferrara · Ferrara, IT

2025 **10th CONCORDi Conference on Corporate R&D and Innovation** – Academic Conference, European Commission Joint Research Centre · Seville, ES

2025 **3rd Economic Fitness and Complexity Summer School** – Summer School, Enrico Fermi Research Centre (9–13 Jun) · Rome, IT

2024 **2nd Economic Fitness and Complexity Summer School** – Summer School, UNU-MERIT (8–12 Jul) · Maastricht, NL

2023 **1st Economic Fitness and Complexity Spring School** – Spring School, Enrico Fermi Research Centre, UNU-MERIT and YSI (5–9 Jun) · Rome, IT

INVITED TALKS AND CONFERENCE PRESENTATIONS (SINCE 2023)

2026 **Online Job Advertisements: Evidence from the data** — Milan, IT
Twin Transition Skill Demand and Regional Productivity

Seminar at Collegio Carlo Alberto – University of Turin — Turin, IT
Skills Demand and Regional Productivity

Workshop discussant – Centro Ricerche Enrico Fermi — Rome, IT
Project Triple T – Tackling a just Twin Transition: a complexity approach to the geography of capabilities, labour markets, and inequalities

International Economic Association World Congress (22–25 Jun) — Belgrade, RS

Regional Studies Association Conference (15–18 Jun) — Gothenburg, SE

Cedefop and Eurostat research conference (27–28 May) — Thessaloniki, GR

8th Geography of Innovation Conference – GEOINNO2026 (28–30 Jan) — Budapest, HU

2025 **UNU-MERIT Development Group seminar** — Online
Assessing Technological Potential through Economic Complexity

TPRI Works-in-Progress Seminar — Online
Scaling Up to the Cloud: Cloud Technology Use and Growth Rates in Small and Large Firms

Productivity Institute online seminar series — Online
Skills demand and Regional Productivity

13th Conference on Micro Evidence on Innovation and Development – MEIDE (10–12 Dec) — Online

Economics of new technology and global change (ENTEG) workshop (17–18 Sep) — Milan, IT

Regional Studies Association Conference (5–9 May) — Porto, PT

2024 **The future of teleworking: The European perspective – Workshop at the London School of Economics** — London, UK
The rebound effects of Teleworking

Guest lecture at the University of Sussex Business School, Science Policy Research Unit — Brighton, UK
Structural Change, Informality and Innovation in sub-Saharan Africa

Guest lecture at the ITU Summer School 'Catalysing the Digital Future: Digital Transformation for Sustainability', UNU-MERIT — Maastricht, NL
What is Economic Complexity?

DRUID Conference (13–15 Jun) — Nice, FR

International Schumpeter Society Conference – ISS2024 (9–11 Jun) — Manchester, UK

ESCoE Conference on Economic Measurement (15–17 May) — Gothenburg, SE

13th Dutch Development Economics Workshop – DuDE (18–19 Apr) — Utrecht, NL

2023 **Panellist at conference in honour of professor Giovanni Dosi, Sant'Anna School of Advanced Studies** — Pisa, IT
On the Shoulders of Giant(s): Charting the Future of Evolutionary Economics

Guest lecture at the University of Sussex Business School, Science Policy Research Unit — Brighton, UK
Structural Change in the Formal and Informal Sectors in sub-Saharan Africa

9th EU Conference on Corporate R&D and Innovation – CONCORDi (24–26 Oct) — Seville, ES

CMD30 – Fismat Conference (4–8 Sep) — Milan, IT

62nd European Regional Science Association (ERSA) Congress (28 Aug – 2 Sep) — Alicante, ES

DRUID Conference (10–12 Jun) — Lisbon, PT

COMPETITIVE GRANTS (SELECTED)

2024 — **The Productivity Institute individual fellowship** — GBP 48,000
Research funding (0.25 FTE) and travel budget for 18 months (until Sep 2025)

2017 — **Doctoral Scholarship (3 years)** — GBP 25,500
Research and teaching scheme stipend and fee waiver from the Science Policy Research Unit of the University of Sussex Business School, Brighton, UK

2014 — **World Wise Web: Capacity Building in the Field of Youth** — EUR 150,000
Key Action 2 Erasmus+ Programme – Capacity Building in the field of Youth. Awarded by EACEA – European Commission with EUR 150,000 funding in 2014 to carry out training activities and fund volunteering programmes to strengthen digital communication capabilities of NGOs active in the field of youth empowerment in Europe, Asia and Latin America

HONORS AND AWARDS

2026 — **JRC Excellence Award**
Awarded for Best Team Collaboration along with JRC Directorate B colleagues who collaborated to organise the B-Days in 2026, which brought together more than 200 colleagues from Seville, Brussels and Ispra to create an inclusive space where staff connected across units, disciplines and professional profiles within Directorate B.

2023 — **Contributi premiali per i ricercatori e assegnisti di ricerca** — EUR 2,000
Research-based award granted by the regional government of Lazio (Italy), funded via the European Social Fund Plus (ESF+) 2021–2027

TRAINING (SELECTED)

2025 — **Causal Inference Workshop** – In-person training sessions by Scott Cunningham, covering: potential outcomes framework, Diff-in-Diff, Shift-Share instruments (3–5 July) · Seville, ES

2023 — **Causal Inference I and II** – Mixtape training sessions by Scott Cunningham, covering: potential outcomes framework and causality fundamentals, DAGs, matching methods, IV, Diff-in-Diff, synthetic control methods (Feb–Mar) · Online

2022 — **Instrumental variables workshop** – Mixtape training sessions by Peter Hull (12–13 Mar) · Online

2021 — **Data Science School on Tools and methods for analysing complex Science, Technology and Innovation (STI) systems** – RISIS Winter School, CNR-IRCrES and AIT (15–26 Feb) · Online

2020 — **Advanced Panel Econometrics: Theory and Applications** – SIdE-Bertinoro Summer School, Italian Econometrics Society (23–29 Aug) · Online